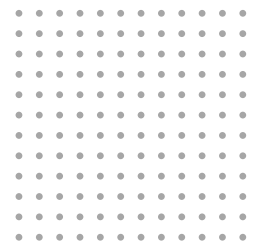




2026 Software Development Internship

Technical Assignment



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1 Overview

Thank you for applying to our internship program. This short technical challenge is designed to evaluate your programming fundamentals, logical thinking, and ability to write clean, understandable code. You are free to use any programming language of your choice.

2 Objective

This assignment is designed to evaluate programming fundamentals, problem solving, code clarity, attention to edge cases, and the ability to explain technical decisions. No specific framework, database, or web technology is required.

Programming language	Any language
Expected effort	Approximately 2-4 hours
Submission deadline	17-9-2026
Submission method	GitHub repository or ZIP file

3 The Challenge: Delivery Route Planner

A delivery company receives a list of delivery requests. Your task is to write a small program that organizes these requests into delivery trips while following the rules below.

3.1 Sample Delivery Data

ID	Area	Priority	Package Weight (kg)
1	Nasr City	2	4.5
2	Maadi	1	2.0
3	Nasr City	3	1.2
4	Zamalek	1	7.0
5	Maadi	2	3.5



Assignment

3.2 Requirements

- Each delivery has an ID, area, priority, and package weight.
- A vehicle can carry a maximum of 10 kg per trip.
- A trip must never exceed the vehicle capacity.
- Deliveries going to the same area should be grouped together where reasonably possible.
- Lower priority numbers represent more urgent deliveries and should be handled first.
- Every valid delivery must appear in exactly one trip.

3.3 Input

Your program should read the deliveries from a file. You may choose CSV, JSON, or a simple text format. Clearly document the format you chose and provide a sample input file with your submission.

3.4 Edge Cases

Your solution should behave sensibly when:

- There are no deliveries.
- A package is heavier than the vehicle's 10 kg capacity.
- Multiple deliveries have the same priority.
- Adding the next package would exceed the vehicle capacity.

There is not necessarily one correct way to handle every edge case. What matters is that your decision is reasonable, your program behaves consistently, and you explain the decision.

4 README and Reasoning

Include a README.md file that explains how to run your program and answers these questions:

1. Explain your solution approach in your own words.
2. What was the most difficult part of the assignment?
3. Are there situations where your algorithm may not produce the best possible grouping? Explain.
4. If the input contained 1,000,000 delivery requests, what part of your solution might become slow or memory-intensive?
5. What would you improve if you had another day to work on the solution?

5 Your Extension

Add one useful feature of your choice that is not explicitly required above. Keep it relevant to the problem and explain briefly in the README why you chose it. The quality of the decision matters more than the size of the feature.



6 Use of AI and Other Resources

You may use documentation, search engines, AI assistants, or other development tools as part of your normal workflow. You do not receive extra credit for avoiding these tools.

You are responsible for understanding every part of your submission.

Candidates who proceed to the next stage will take part in a short technical discussion. You may be asked to explain design decisions, trace the execution of your program, identify a problem in your code, or make a small change to your own solution.

7 Submission Requirements

- Source code for the complete solution.
- A README.md containing setup/run instructions and the reasoning questions above.
- At least one sample input file that can be used to run the program.
- Submit through a GitHub repository. If this is not possible, a ZIP file is acceptable.
- Send your solution through HR@et3.co

8 What We Are Looking For

This is an internship challenge, not a production-system assignment. We are not looking for advanced frameworks, complicated architecture, or clever code for its own sake. We are looking for evidence that you can understand a problem, break it down, write correct and readable code, think about unusual cases, and explain your decisions.

A simple solution that you fully understand is preferable to a sophisticated solution that you cannot explain.

9 Submission Checklist

- Program runs successfully.
- Sample input is included.
- README explains how to run the program.
- README answers all five reasoning questions.
- One additional feature is included and explained.
- Submission is sent before the deadline.

10 Deadline

Please submit your solution by **Thursday 17 Sep 2026 11:59 pm**. If you need more time or have any questions, let us know.

11 Company Information

11.1 Company Profile

[Click here to view the company profile.](#)

11.2 Company Projects Portfolio

[Click here to view the company projects portfolio.](#)